



Degree project presentation

University West

Trollhättan, Sweden

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Title

Evaluation of Whole-Body Control Architectures for Humanoid Balance in Simulation with Learning-Based Residual Compensation

Sanju N Sebastian

University West

Master in Robotic and Automation Degree project, 30 ECTS credits

University West, Dep of Production System

Trollhättan, Sweden

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Introduction

Humanoid balance failure is an unsolved problem

Real incidents (2024–2025):

- Unitree H1 and G1 fell publicly at multiple demos
- At CES 2024, every robot had human spotters within arm's reach — engineering teams knew they could fall without warning
- AIDOL (Russia, Nov 2025) collapsed on stage in front of 50 journalists

Industry expert quote:

"They're not very good at recovery. These are heavy machines. If they fall onto a person, a person could be seriously injured." — Roberta Nelson Shea, Universal Robotics Safety Pioneer, 2025

Source: Automate.org Industry Insights, 2025 -

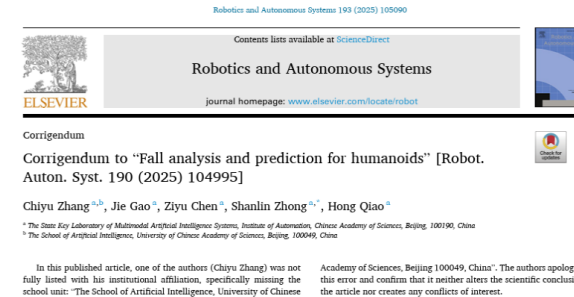
<https://www.automate.org/industry-insights/humanoid-robots-are-evolving>

This is not just a demo problem — fall prevention in bipedal robots is an active research area with papers published in 2025 specifically on this challenge.



AIDOL humanoid robot collapses at Moscow tech event, November 11, 2025

Source: Euronews / Moskva City News Agency via Reuters, November 2025



Corrigendum

Corrigendum to "Fall analysis and prediction for humanoids" [Robot. Auton. Syst. 190 (2025) 104995]

Chiyu Zhang^{a,*}, Jie Gao^a, Ziyu Chen^a, Shanlin Zhong^{a,b}, Hong Qiao^a

^a The State Key Laboratory of Multimodal Artificial Intelligence Systems, Institute of Automation, Chinese Academy of Sciences, Beijing, 100190, China

^b The School of Artificial Intelligence, University of Chinese Academy of Sciences, Beijing, 100049, China

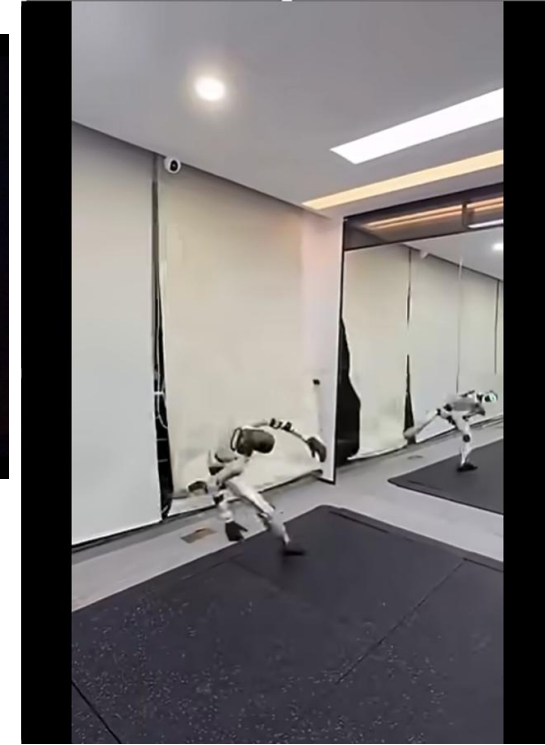
In this published article, one of the authors (Chiyu Zhang) was not fully listed with his institutional affiliation, specifically missing the school unit: "The School of Artificial Intelligence, University of Chinese Academy of Sciences, Beijing 100049, China". The authors apologize for this error and confirm that it neither alters the scientific conclusions of the article nor creates any conflicts of interest.

A peer-reviewed paper in Robotics and Autonomous Systems (2025) specifically about fall prediction in humanoid robots.

Credit: Bouazza et al. (2025). Fall analysis and prediction for humanoids. Robotics and Autonomous Systems, 188, 104956.

Source:

<https://www.sciencedirect.com/science/article/abs/pii/S0921889025000818>



Unitree humanoid robot loses balance during movement

Public demonstration, 2025

Source: YouTube, 2025. Available at:

<https://www.youtube.com/shorts/wOaKbpFNIVI>

Introduction

WBC computes joint torques satisfying multiple objectives simultaneously — several architectures exist but no unified push-recovery comparison existed.

WBC exists — but which architecture is best?

- Whole-body control (WBC) computes joint torques satisfying multiple objectives simultaneously [Escande et al. 2014, Koolen et al. 2016]
- Four architectures used in literature: WQP, HQP, Passivity-Based, LQR-on-LIPM
- Recent paper (Sovukluk et al. 2025, arXiv) compares only ID-WBC vs Passivity — no unified four-way comparison exists —
- Frontiers benchmarking (2022) compared WBC on TALOS robot — but no push-recovery evaluation, no residual learning

Gap: **No unified push-recovery comparison of all four architectures with statistical grounding exists**

No prior work has evaluated all four WBC architectures together in a push-recovery context with statistical grounding.

Study	WQP	HQP	Passivity	LQR	Push recovery
Escande 2014	✓	✓	✗	✗	✗
Sovukluk 2025	✗	✗	✓	✗	✗
Frontiers 2022	✓	✗	✗	✗	✗
THIS THESIS	✓	✓	✓	✓	✓

This thesis fills that gap — first unified four-way comparison with 200-seed stochastic evaluation and McNemar statistical validation.

Experimental Comparison of Whole-Body Control Formulations for Humanoid Robots in Task Acceleration and Task Force Spaces

Sait Sovukluk^{1,1*}, Grazia Zambella^{1,1*}, Tobias Egle¹, and Christian Ott^{1,2}

Abstract—This paper studies the experimental comparison of two different whole-body control formulations for humanoid robots: inverse dynamics whole-body control (ID-WBC) and passivity-based whole-body control (PB-WBC). The two controllers fundamentally differ from each other as the first is formulated in task acceleration space and the latter is in task force space with passivity considerations. Even though both control methods predict stability under ideal conditions in closed-loop dynamics, their robustness against joint friction, sensor noise, unmodeled external disturbances, and non-perfect contact conditions is not evident. Therefore, we analyze and experimentally compare the two controllers on a humanoid robot platform through swing foot position and orientation control, squatting with and without unmodeled additional weights, and jumping. We also relate the observed performance and characteristic differences with the controller formulations and highlight each controller's advantages and disadvantages.

I. INTRODUCTION

Humanoid robots are multi-purpose complex systems that are designed to imitate human appearance and behavior [1], [2]. They are designed to perform human labor tasks, such as carrying weights, manipulating the environment, performing high-risk tasks, and assisting/supporting humans in their environments. The DARPA Robotics Challenge in 2015 summarizes the best usage of such robots [3], [4]. The challenge required humanoid robots to replace humans in a dangerous rescue mission. This mission included driving a golf cart; walking through doors, uneven/unorganized surfaces, and stairs; using power tools such as drillers and cutters; and manipulating the environment such as opening/closing valves, replacing items, plugging in power cords, and opening doors. Achieving more than 75 degrees of freedom (DoF) per



Fig. 1. Snapshots of experiments covered in this paper: jumping (top), disturbed and undisturbed squatting (bottom left), and foot position and orientation control (bottom right). Supplemental video collects all experiments.

Most recent WBC comparison (Sovukluk et al. 2025) — evaluates only 2 of the 4 architectures

Credit: **Sovukluk et al. (2025). Experimental Comparison of Whole-Body Control Formulations for Humanoid Robots.**
arXiv:2507.18502

Source: <https://arxiv.org/abs/2507.18502>



Aims

Research aims

- **Aim 1:** Evaluate and compare four WBC architectures for humanoid push-recovery balance under stochastic disturbances
- **Aim 2:** Determine whether a learned residual MLP can improve robustness beyond model-based control alone

Research Questions

- **RQ1:** Which WBC architecture achieves the highest push-recovery stability under stochastic disturbances?
- **RQ2:** Do WQP and HQP produce meaningfully different outcomes despite architectural differences?
- **RQ3:** Can a learned residual MLP improve robustness beyond the best model-based controller alone?
- **RQ4:** Does residual learning generalise across all four WBC architectures?

Controllers evaluated: Weighted QP · Hierarchical QP · Passivity-Based · LQR-on-LIPM

Background

Why these four controllers?

These four represent the complete design space of model-based WBC:

- **WQP**
 - All objectives as weighted sum
 - Industry standard Used by Boston Dynamics, Tsinghua CA-WBC (2024) → Baseline for all modern WBC
- **HQP**
 - Strict priority order higher never violated
 - Safety-critical variant Used in surgical robots, space robotics [Escande et al. 2014] → Tests whether strict priority adds value
- **Passivity-Based**
 - Energy dissipation guarantees stability
 - Energy-shaped stability Used in KUKA iiwa, compliant interaction [Sovukluk et al. 2025] → Tests whether energy dissipation outperforms optimisation
- **LQR-on-LIPM**
 - Simplified inverted pendulum model
 - Reduced-order baseline Foundation of ASIMO, early humanoid [Kajita et al. 2001] → Tests cost of model simplification
- **Together: complete spectrum from simple to complex**
- **Key gap in literature: No unified comparative evaluation existed**

Hierarchical Quadratic Programming Part 1: Fundamental Bases

Adrien Escande
JRL-CNRS/ARIST
Toulouse, Japan

Nicolas Mansard
LAAS-CNRS, Univ. Toulouse
Toulouse, France

Pierre-Brice Wieber
INRIA Grenoble
St Ismier, France

Abstract

Hierarchical least-square optimization is often used in robotics to inverse a direct function when multiple incompatible objectives are involved. Typical examples are inverse kinematics or dynamics. The objectives can be given as equalities to be satisfied (e.g. point-to-point task) or as areas of satisfaction (e.g. the joint range). This two-part paper proposes a complete solution to resolve multiple least-square quadratic problems of both equality and inequality constraints ordered into a strict hierarchy. Our method is able to solve a hierarchy of only equalities ten times faster than the classical method and can consider inequalities at any level while running at the typical control frequency on whole-body size problems. This generic solver is used to resolve the redundancy of humanoid robots while generating complex movements in constrained environment.

In this first part, we establish the mathematical bases underlying the hierarchical problem and propose a dedicated solver. When only equalities are involved, the solver amounts to the classical solution used to handle redundancy in inverse kinematics in a far more efficient way. It is able to handle inequalities at any priority level into a single resolution scheme, which avoids the high number of iterations encountered with cascades of solvers. A simple example is given to illustrate the interest of our approach. The second part of the paper focuses on the implementation of the solver for inverse kinematics

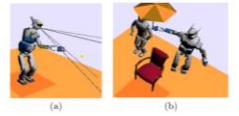


Fig. 1: Various situations of inequality and equality constraints. (a) Reaching a distant object while keeping balance. The visibility and postural tasks are satisfied only if possible. (b) The target is kept inside a preference area.

1 Introduction

Least squares are a mean to satisfy at least a set of constraints that may not be feasible. When the constraints are linear, the least squares are written as a quadratic program, whose solution is given for example by the pseudo-inverse. Linear least squares have been widely used in robot control in the frame of instantaneous task resolution [De Schutter and Van Brussel, 1988], inverse kinematics [Whitney, 1972] or operational-space inverse dynamics [Khatib, 1987]. These approaches describe the objectives to be pursued by the robot using a function depending on the robot configuration.

Escande et al. (2014) — foundational HQP paper
Source: IEEE Transactions on Robotics, 2014

frontiers
in Robotics and AI

ORIGINAL RESEARCH
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Benchmarking Whole-Body Controllers on the TALOS Humanoid Robot

N. Ramazzat^{1,2}, O. Stasse^{1*} and S. Boria²

¹LAAS-CNRS, University of Toulouse, Toulouse, France; ²Artibus, Toulouse, France; ³Artificial and Natural Intelligence, Toulouse Institute, Toulouse, France

This article presents a comparison of three control schemes applied on the commercially available TALOS humanoid robot. The aim is to highlight the advantages and drawbacks of each model applied on three locomotion problems: walking on flat and non-flat terrain and climbing stairs. The different models are based on position control (first and second models) or torque control (third model). The first one uses a hierarchical quadratic program at velocity level. The second one uses a weighted quadratic program named Task Space Inverse Dynamic (TSID) at acceleration level. Finally, the last one also uses TSID but at torque level. The controller performances are compared in simulation, using Gazebo, on the accuracy of their tracking, their energy consumption, and their computational time execution.

Keywords: humanoid robot, benchmarking, whole body control, balance, torque control

1 INTRODUCTION

1.1 Goal

Bipedal locomotion of humanoid robots is considered a difficult problem because of the complexity of robot dynamics, the numerous constraints of the motion, and the unknown environment. The design choice made when designing a robot may have a strong impact on the control laws that are really working on the system and the real performances. A recent example is the Digit robot which has very impressive capabilities by choosing a careful tradeoff between the chosen actuation technology and robot weight distribution [Robotics, 2022]. The robot is very robust to impact and allows torque control but is slightly limited by the payload it can carry (10 kg). Realizing torque control on electric-based bipedal system is challenging. If it was successfully realized on the TORO robot [Engelbrecht et al., 2014] for standing whole body control and walking, it is notoriously more difficult to achieve than position control. A striking example is given by the

Caption: WBC benchmarking on TALOS robot - no push-recovery evaluation included

Credit: Frontiers in Robotics and AI, 2022

Background

Why add residual learning?

- Model-based WBC has a ceiling:
 - All controllers hit ankle saturation above 175 N
 - Model errors accumulate under large disturbances
 - Full model knowledge is never available in practice
- Residual learning closes the gap:
 - Johannink et al. 2019: residual RL on top of model-based controllers improves robustness
 - Kasaei et al. 2023: largest gains in mid-force recovery regime (125–175 N)
- Gap in prior work: Only tested on single controllers — generalisation across architectures was unknown
- Contribution: **First systematic residual learning evaluation across four WBC architectures**

Residual Reinforcement Learning for Robot Control

Tobias Johannink^{*1,3}, Shikhar Bahl^{*2}, Ashvin Nair^{*2}, Jianlan Luo^{1,2}, Avinash Kumar¹, Matthias Loskyll¹, Juan Aparicio Ojea¹, Eugen Solowjow¹, Sergey Levine²

Abstract—Conventional feedback control methods can solve various types of robot control problems very efficiently by capturing the structure with explicit models, such as rigid body equations of motion. However, many control problems in modern manufacturing deal with contacts and friction, which are difficult to capture with first-order physical modeling. Hence, applying control design methodologies to these kinds of problems often results in brittle and inaccurate controllers, which have to be manually tuned for deployment. Reinforcement learning (RL) methods have been demonstrated to be capable of learning continuous robot controllers from interactions with the environment, even for problems that include friction and contacts. In this paper, we study how we can solve difficult control problems in the real world by decomposing them into a part that is solved efficiently by conventional feedback control methods, and the residual which is solved with RL. The final control policy is a superposition of both control signals. We demonstrate our approach by training an agent to successfully perform a real-world block assembly task involving contacts and unstable objects.

1. INTRODUCTION

Robots in today's manufacturing environments typically perform repetitive tasks, and often lack the ability to handle variability and uncertainty. Commonly used control algorithms, such as PID regulators and the computed torque method, usually follow predefined trajectories with little adaptive behavior. Many manufacturing tasks require some degree of adaptability or feedback to the environment, but significant engineering effort and expertise is required to design feedback control algorithms for these industrial robots. The engineering time for fine tuning such a controller might be similar in cost to the robot hardware itself. Being able to quickly and easily design feedback controllers for industrial robots would significantly broaden the space of manufacturing tasks that can be automated by robots.

Why is designing a feedback controller for many tasks hard with classical methods? While conventional feedback control methods can solve tasks such as path following efficiently, applications that involve contacts between the robot and its environment are difficult to approach with conventional control methods. Identifying and characterizing

behavior generalization. Thus, all possible system behaviors must be considered a priori at design time.

Reinforcement learning (RL) methods hold the promise of solving these challenges because they allow agents to learn behaviors through interaction with their surrounding environments and ideally generalize to new scenarios that differ from the specifications at the control design stage. Moreover, RL can handle control problems that are difficult to approach with conventional controllers because the control goal can be specified indirectly as a term in a reward function and not explicitly as the result of a control action. All of these aspects are considered enablers for truly autonomous manufacturing systems and important for fully flexible lot-size one manufacturing [1]. However, standard RL methods require the robot learn through interaction, which can be unsafe initially, and collecting the amount of interaction that is needed to learn a complex skill from scratch can be time consuming.

In this paper, we study control problems that are difficult to approach with conventional feedback control methods. However, the problems possess structure that can be partially handled with conventional feedback control, e.g. with impedance control. The residual part of the control task, which is the part that must consider contacts and external object dynamics, is solved with RL. The outputs of the conventional controller and RL are superposed to form the commanded control. The main contribution of this paper is a methodology that combines conventional feedback control with deep RL methods and is illustrated in Fig. 1. Our main motivation is a control approach that is suitable for real-world control problems in manufacturing, where the exploratory behavior of RL is a safety concern and the data requirements of deep RL can be expensive. We provide a thorough evaluation of our method on a block assembly task in simulation and on physical hardware. When the initial orientation of the blocks is noisy, our hand-designed controller fails to solve the task, while residual RL successfully learns to perform the task in under 3 hours. This suggests that we could usefully apply our method to practical manufacturing problems.

Caption: Foundational residual learning paper

Credit: Johannink et al. (2019). ICRA 2019.

IEEE. DOI: 10.1109/ICRA.2019.8794127

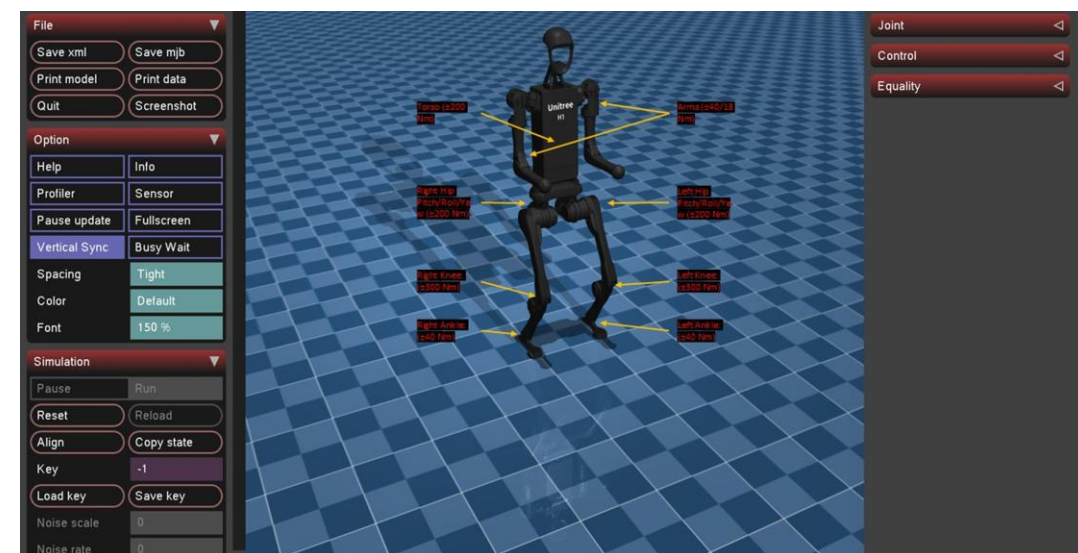
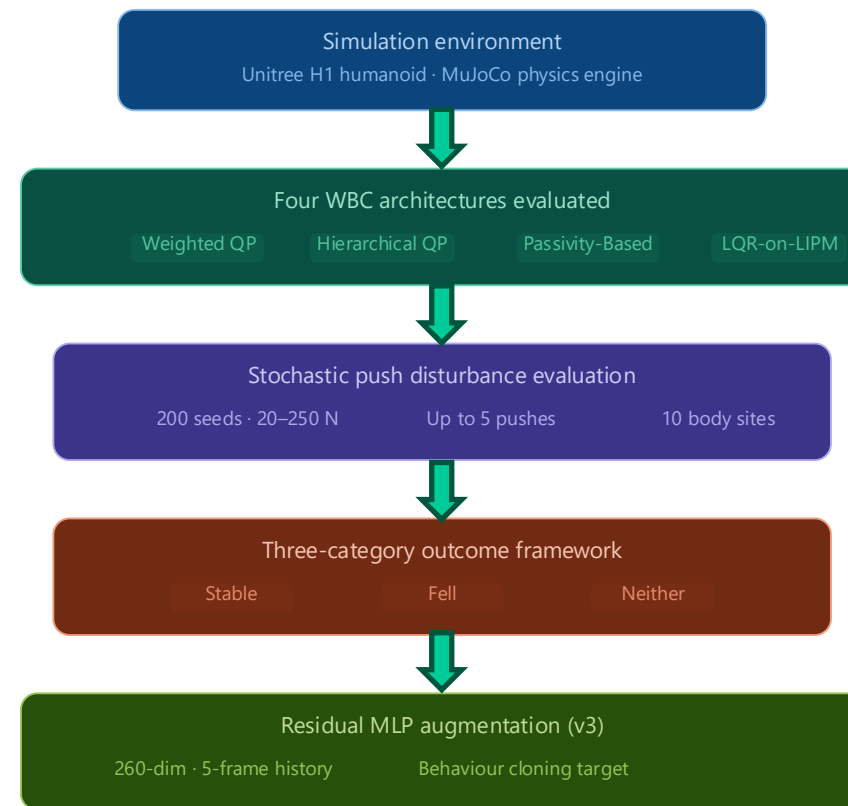
$$\begin{array}{ccccc} \text{Model-based WBC} & & \text{Residual MLP} & & \\ \tau_{\text{base}} & + & \Delta\tau & = & \tau_{\text{final}} \\ (\text{passivity}) & & (\text{learned}) & & (\text{improved}) \end{array}$$

[Johannink et al. 2019 — foundational concept]

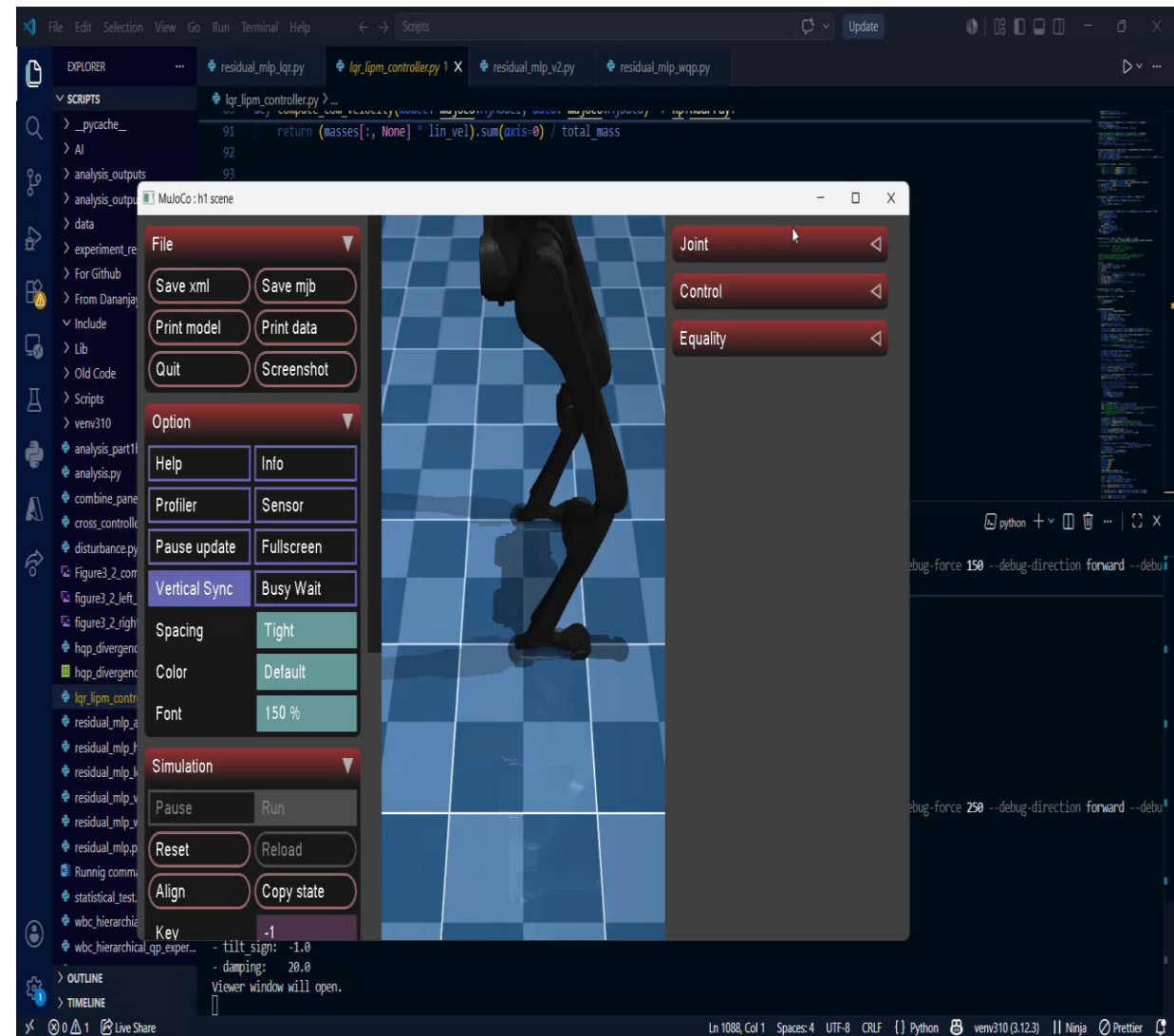
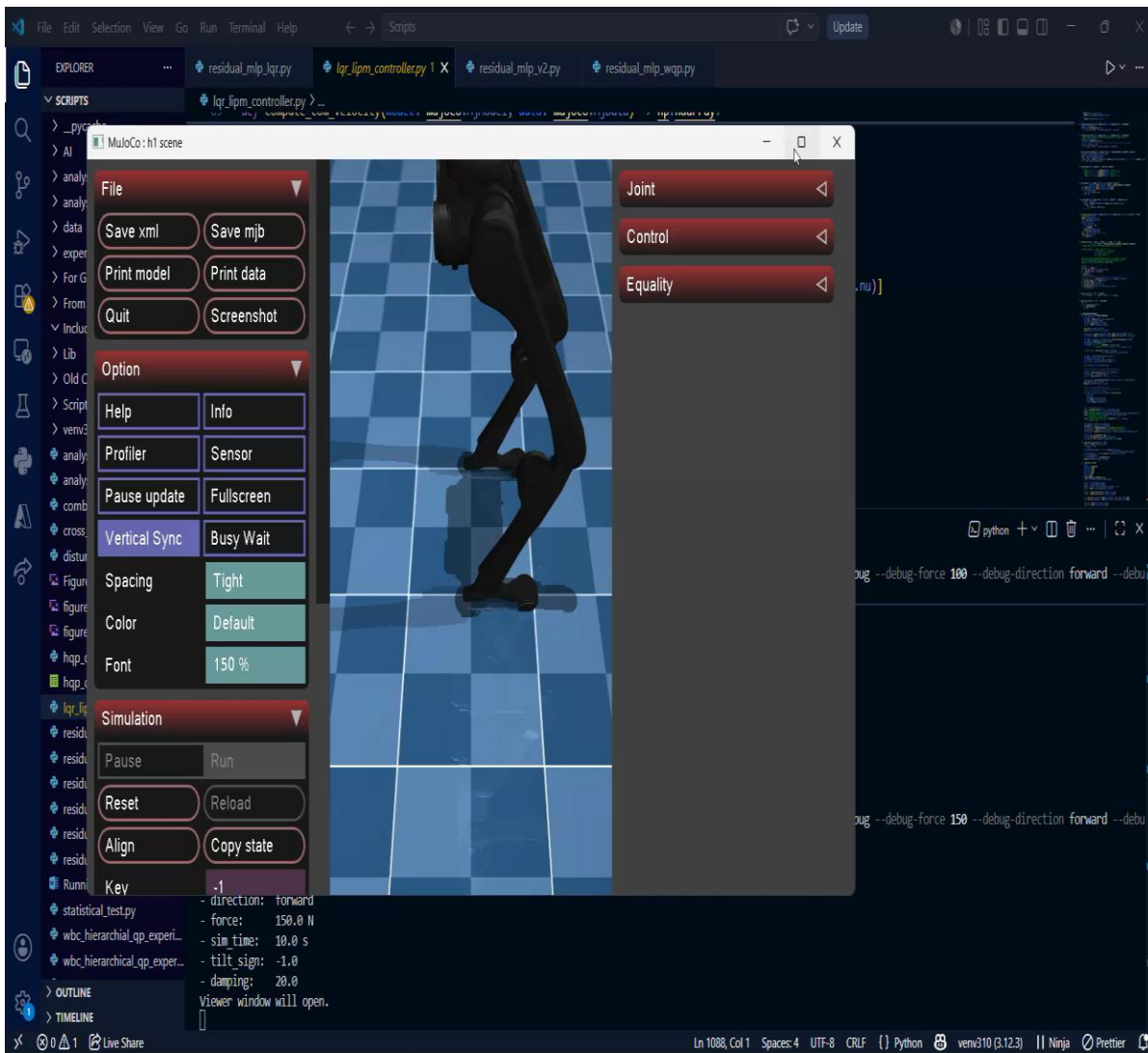
Methodology

Experimental methodology

- Robot platform: **Unitree H1** humanoid · **MuJoCo** simulation
 - Standard in humanoid WBC research [Todorov et al. 2012 — MuJoCo original paper]
 - Unitree H1 model validated against hardware
 - Allows $200 \times 4 = 800$ controlled experiments impossible on hardware
- Disturbance protocol LogUniform 20–250 N · 10 body sites Up to 5 sequential pushes · 200 seeds per controller
 - LogUniform ensures proportional coverage across force magnitudes
 - 200 seeds gives ~70% statistical power at $\alpha=0.05$ for 5–6pp differences
 - Above 250 N: all controllers saturate — no discriminating information
- Outcome framework (novel contribution) Stable / Fell / Neither → Binary frameworks miss the "neither" category
 - Binary frameworks used in prior work miss lingering/oscillating states
 - Novel contribution — 7% of trials fall in this category for WQP/HQP
- Statistical validation McNemar's exact test on paired 200-seed outcomes
 - Seeds are paired — same disturbance, two controllers — requires paired test
 - Standard chi-squared assumes independence which is violated here



Methodology: What Simulation Looks Like



Approach

Phase 1 — Controller Implementation

Four controllers implemented in Python + MuJoCo:

- WQP/HQP: QP solver (quadprog) at each timestep
- Passivity: damping injection — zero solver overhead
- LQR: offline Riccati solution — matrix multiply only

Why these implementations?

- Each reflects the theoretical design of the architecture
- Same simulation environment ensures fair comparison
- No confounds from different platforms or sensors

Validation — HQP divergence test:

- At 50:50 weighting: 200 Nm max difference confirmed
- Hierarchy is real and correctly implemented
- Equivalence at 2000:1 is a finding, not a bug

Phase 2 — Identifying the ceiling

Running all four controllers revealed a hard limit: All controllers converge to 16% above 175 N Ankle saturation — model-based control cannot overcome this alone This motivated the residual MLP

Phase 3 — Residual MLP development

v1: 42-dim input, LIPM loss —→ NULL
v2: 46-dim input, LIPM loss —→ NULL

Root cause identified:

LIPM eigenvalue = 1.003 → gradient ≈ 0 everywhere
±50 Nm correction → less than 0.3% loss change
Numerically verified — not a tuning problem

v3: 260-dim, BC target, 5-frame history → POSITIVE

Three fixes that made v3 work: -

Ankle strategy BC target — non-zero, physically motivated
5-frame history — detects disturbance events
τ_base in input — complements base controller

MLP Residual Learning — Complete Version Progression (All Controllers)

Version	Base	Input dim	Target	History	Stable/200	vs Baseline
v1	Passivity	42	Zero (LIPM loss)	No	109/200 (54.5%)	-0.5%
v2	Passivity	46	Zero (LIPM loss)	No	109/200 (54.5%)	-0.5%
v3	Passivity	260	Pitch-rate BC	Yes (5 fr.)	125/200 (62.5%)	+7.5pp
v3	WQP	260	Pitch-rate BC	Yes (5 fr.)	114/200 (57.0%)	+5.5pp
v3	HQP	260	Pitch-rate BC	Yes (5 fr.)	114/200 (57.0%)	+5.5pp
v3	LQR	260	Pitch-rate BC	Yes (5 fr.)	120/200 (60.0%)	+2.5pp
Distillation	WQP→Pass.	260	Passivity torques	Yes (5 fr.)	125/200 (62.5%)	+11.0pp

MLP architecture evolution — three versions tested. V1 and V2 confirmed null. V3 confirmed positive.

Results

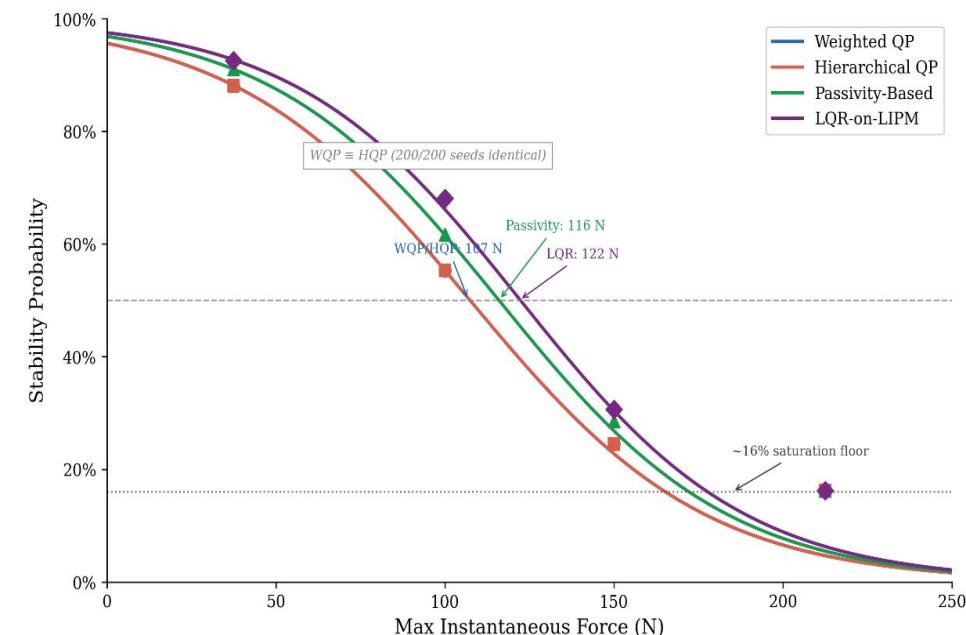
Base controller results (200 seeds each)

Controller	Stable	Crossover	CoM displacement
WQP	51.5%	~107 N	0.0410 m
HQP	51.5%	~107 N	0.0410 m
Passivity	55.0%	~116 N	0.0365 m
LQR	57.5%	~122 N	0.0410 m

← identical

← best quality

← highest rate

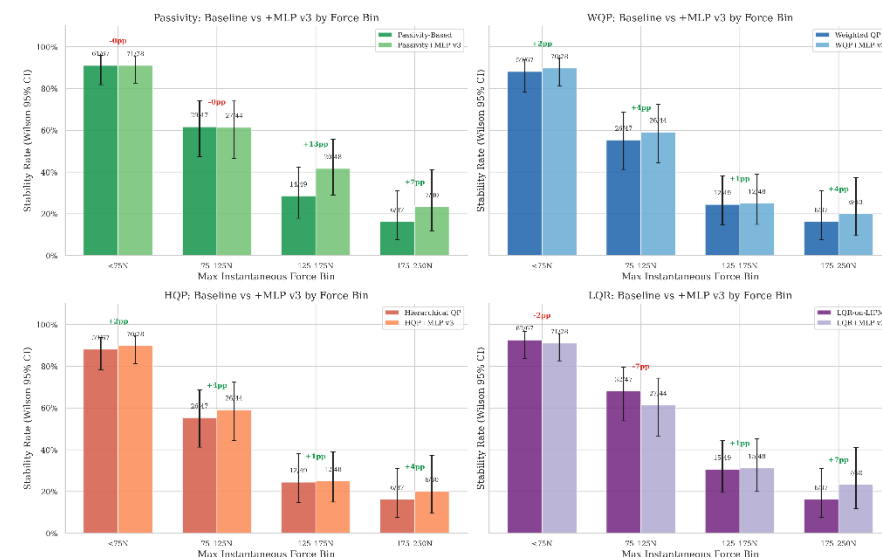


Stability rate vs push force magnitude —
sigmoid crossover points per controller
Source: Own experimental results, April 2026

Key finding:

- WQP = HQP on all 200 seeds
- HQP costs 1.89× more compute for zero benefit
- Passivity: 11% lower CoM displacement
- Neither category: 7% WQP/HQP vs 4.5% & Passivity/LQR Binary frameworks miss this entirely
- All controllers hit 16% floor above 175 N - Ankle saturation — physical limit, not learnable

MLP v3 Force-Bin Breakdown — All Four Base Controllers



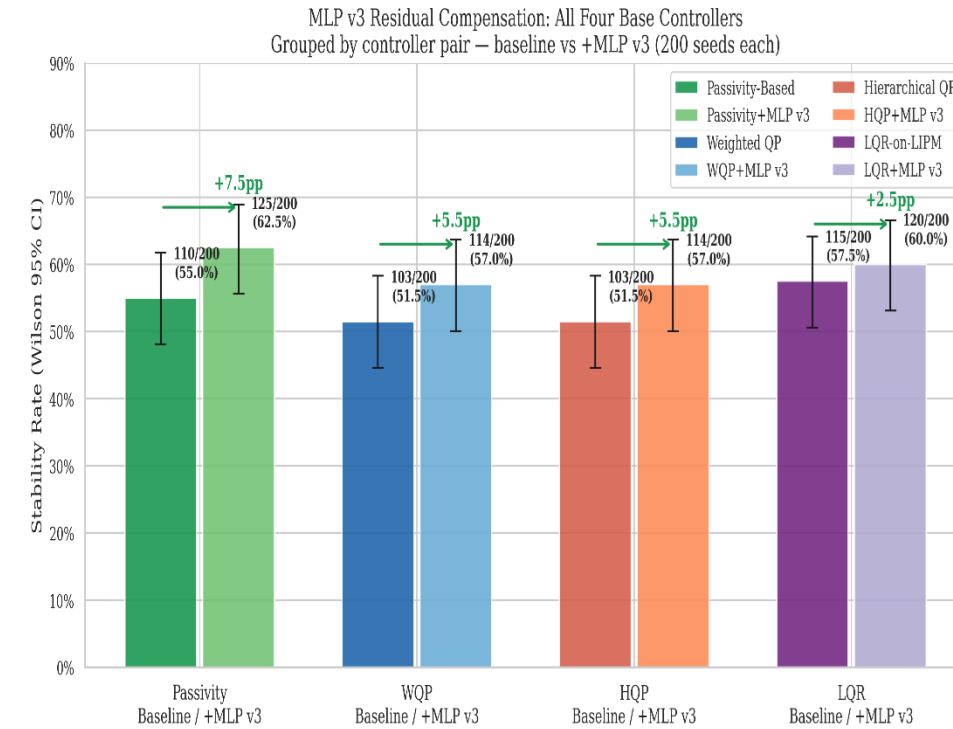
Results

MLP v3 results across all controllers

Controller	Baseline	+MLP v3	Gain
Passivity	55.0%	62.5%	+7.5pp
WQP	51.5%	57.0%	+5.5pp
HQP	51.5%	57.0%	+5.5pp
LQR	57.5%	60.0%	+2.5pp

←primary

- $p = 3.1 \times 10^{-5}$ (McNemar exact test)
- $b = 0$ — never destabilised a stable trial
- 15 additional trials recovered
- Largest gain at 125–175 N (+13pp) — exactly at feasibility boundary
- Improvement scales with error structure not controller complexity



MLP v3 residual compensation across all four base controllers (200 seeds each, Wilson 95% CI)

Discussion

What the results mean

- WQP $\equiv$ HQP finding
 - Strict hierarchy adds no value when objectives don't genuinely conflict.
 - At 2000:1 balance dominance the hierarchy is never exercised. Don't use HQP for balance alone.
- Passivity vs LQR tradeoff
 - LQR wins at low force — simplified model sufficient
 - Passivity wins on motion quality across all regimes
 - Choice depends on operational requirement
- Why MLP v3 succeeded where v1/v2 failed
 - Physically grounded target + temporal context
 - Loss function design matters more than network architecture
 - Largest gain at 125–175 N — feasibility boundary
- Remaining limitation
 - 175 N floor unchanged — ankle saturation is a physical hardware constraint, not a learning problem. Requires stepping controller.

Engineering ethics & sustainability

- Safety imperative
 - Falls cause serious injury — 89 kg robot falling on a human is a safety-critical event
 - This work contributes to the safety foundation required before deployment in shared spaces
- Simulation-first principle
 - All experiments in simulation — no risk to physical hardware or human subjects
 - Reproducible: fixed seeds, open methodology
- Sustainability angle
 - Passivity-based control uses 0ms solver overhead vs 0.84ms (WQP) and 1.58ms (HQP) per step
 - At 500Hz control: HQP wastes $0.74\text{ms} \times 500 = 370\text{ms}$ of compute per second vs passivity
 - Lower compute = lower energy consumption at scale across robot fleets
- Societal relevance
 - Robust balance is prerequisite for humanoid deployment in hospitals, elder care, industry
 - Understanding architectural tradeoffs prevents dangerous deployment choices

Conclusion

1. WQP = HQP — strict hierarchy adds no benefit at 2000:1 balance dominance (1.89× compute wasted)
2. Passivity provides best CoM quality — 11% lower displacement, fewest lingering outcomes
3. LQR quantifies reduced-order modelling cost — accurate in linear regime, blind above 175 N
4. Residual learning requires physically grounded targets and temporal context to succeed
5. MLP v3 generalises across all four controllers — improvement scales with error structure

The thesis title is fully delivered:

WBC architecture evaluation

Learning-based residual compensation

Future work

- Immediate extensions
 - RL fine-tuning for true residual labels (supervised BC is an approximation)
 - Larger history window — LSTM instead of frame stacking
 - Hardware validation on physical Unitree H1
- Broader directions
 - Stepping controller for high-force regime (deliberately out of scope here)
 - MPC integration for predictive recovery
 - Multi-contact scenarios beyond static balance Industrial relevance
 - Framework directly applicable to any WBC deployment requiring push-recovery validation

Key References

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Thanks for your interest!

Sanju N Sebastian